

ALGORITHM AND DATASET – SALES PERFORMANCE DASHBOARD

1. Algorithm

1.1 Step-by-Step Procedure

1. Load the Superstore sales dataset using Python and Pandas.
2. Inspect the dataset for records, columns, data types, missing values, and duplicate records.
3. Remove duplicate records if present.
4. Convert Order Date and Ship Date into datetime format.
5. Standardize column names.
6. Create derived features such as Order Year, Order Month, Order Quarter, Delivery Days, Profit Margin, and Sales Per Unit.
7. Perform Exploratory Data Analysis (EDA) on sales, profit, quantity, discount, category, sub-category, region, and time.
8. Calculate key performance indicators such as Total Sales, Total Profit, Total Orders, Total Quantity, and Profit Margin.
9. Aggregate sales and profit by category, sub-category, region, product, and time period.
10. Generate suitable visualizations from the analyzed data.
11. Present the results through the Sales Performance Dashboard.

1.2 Logic / Method Used

Data Cleaning

- Missing-value checking
- Duplicate detection
- Data-type conversion
- Column standardization

Data Transformation

- Date-based feature extraction
- Feature engineering
- Profit Margin calculation

- Sales Per Unit calculation

Data Analysis

- Group-by aggregation
- Comparative analysis
- Time-based analysis
- Category and regional analysis

KPI Calculation

- Total Sales = Sum of Sales
- Total Profit = Sum of Profit
- Total Quantity = Sum of Quantity
- Total Orders = Number of Unique Order IDs
- Profit Margin = $(\text{Total Profit} / \text{Total Sales}) \times 100$
- Sales Per Unit = Sales / Quantity

1.3 Model / Technique Applied

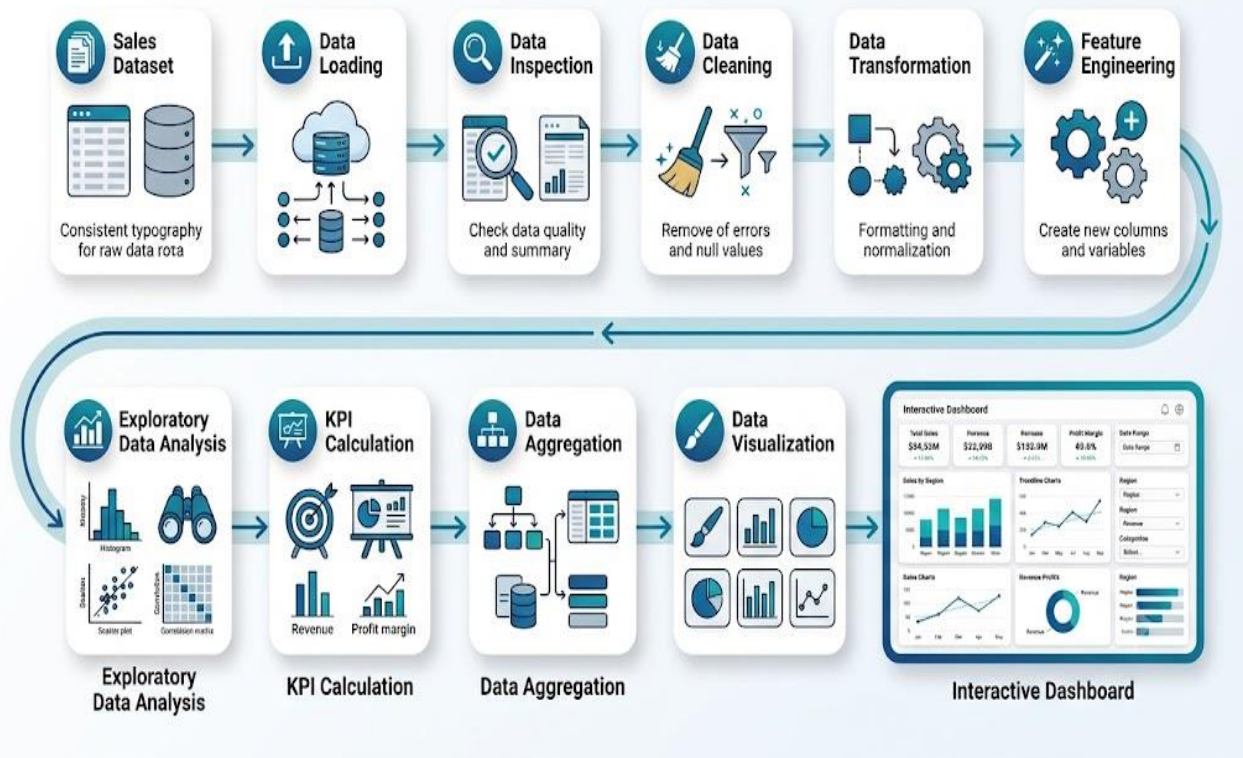
The project uses:

- Exploratory Data Analysis (EDA)
- Descriptive Data Analytics
- Data Preprocessing
- Feature Engineering
- Data Aggregation
- KPI Analysis
- Trend Analysis
- Data Visualization

Machine-learning prediction models are not used in this project.

1.4 Process Flow

Sales Performance Dashboard



2. Dataset

2.1 Source of Data

Dataset Name: Sample Superstore Dataset

Source: Kaggle – Superstore Dataset Final

Dataset Link:

<https://shorturl.at/AbjRa>

The dataset contains retail sales transaction data used for sales and business analytics.

2.2 Number of Records / Samples

Total Records: 9,994

Total Features / Attributes: 21

2.3 Features / Attributes

No.	Attribute	Description	Data Type
1	Row ID	Unique record identifier	Integer
2	Order ID	Unique order identifier	String
3	Order Date	Date of order	Date
4	Ship Date	Date of shipment	Date
5	Ship Mode	Shipping method	Categorical
6	Customer ID	Unique customer identifier	String
7	Customer Name	Customer name	Categorical
8	Segment	Customer segment	Categorical
9	Country	Customer country	Categorical
10	City	Customer city	Categorical
11	State	Customer state	Categorical
12	Postal Code	Customer postal code	Numerical
13	Region	Sales region	Categorical
14	Product ID	Unique product identifier	String
15	Category	Product category	Categorical
16	Sub-Category	Product sub-category	Categorical
17	Product Name	Product name	Categorical
18	Sales	Sales amount	Numerical
19	Quantity	Quantity sold	Numerical
20	Discount	Discount applied	Numerical
21	Profit	Profit generated	Numerical

2.4 Training and Testing Data Split

Training Data: Not Applicable

Testing Data: Not Applicable

Reason: This project uses descriptive and exploratory data analytics and does not use a machine-learning prediction model. Therefore, the complete dataset is used for analysis instead of dividing it into training and testing datasets.

2.5 Data Format / Type

File Format: CSV

Data Structure: Tabular

Records: 9,994

Attributes: 21

Data Types:

- **Numerical:** Row ID, Postal Code, Sales, Quantity, Discount, Profit
- **Categorical/Text:** Order ID, Ship Mode, Customer ID, Customer Name, Segment, Country, City, State, Region, Product ID, Category, Sub-Category, Product Name
- **Date:** Order Date, Ship Date

3. Implementation Code

```
import pandas as pd
import numpy as np

df = pd.read_csv("Sample - Superstore.csv")

print("Dataset Shape:", df.shape)
print(df.head())
print(df.info())

print("\nMissing Values:")
print(df.isnull().sum())
```

```
print("\nDuplicate Records:", df.duplicated().sum())
```

```
df = df.drop_duplicates()
```

```
df.columns = (  
    df.columns  
    .str.strip()  
    .str.lower()  
    .str.replace(" ", "_")  
)
```

```
df["order_date"] = pd.to_datetime(df["order_date"])
```

```
df["ship_date"] = pd.to_datetime(df["ship_date"])
```

```
df["order_year"] = df["order_date"].dt.year
```

```
df["order_month"] = df["order_date"].dt.month
```

```
df["order_quarter"] = df["order_date"].dt.quarter
```

```
df["delivery_days"] = (  
    df["ship_date"] - df["order_date"]  
) .dt.days
```

```
df["profit_margin"] = (  
    df["profit"] / df["sales"]  
) * 100
```

```
df["sales_per_unit"] = (  
    df["sales"] / df["quantity"]  
)
```

```
total_sales = df["sales"].sum()
total_profit = df["profit"].sum()
total_quantity = df["quantity"].sum()
total_orders = df["order_id"].nunique()
```

```
profit_margin = (
    total_profit / total_sales
) * 100
```

```
category_sales = (
    df.groupby("category")["sales"]
    .sum()
    .sort_values(ascending=False)
)
```

```
region_sales = (
    df.groupby("region")["sales"]
    .sum()
    .sort_values(ascending=False)
)
```

```
monthly_sales = (
    df.groupby("order_month")["sales"]
    .sum()
)
```

```
print("\nTotal Sales:", total_sales)
print("Total Profit:", total_profit)
```

```
print("Total Quantity:", total_quantity)
print("Total Orders:", total_orders)
print("Profit Margin:", profit_margin)
```

```
print("\nSales by Category:")
print(category_sales)
```

```
print("\nSales by Region:")
print(region_sales)
```

```
print("\nMonthly Sales:")
print(monthly_sales)
```